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Lecture 6 Summary: Provable Guarantees for Policy Gradient Methods

Overview

This lecture studies the theoretical foundations and convergence guarantees of policy gradient methods, based on *Kakade & Langford (2002)* and *Schulman et al. (2015)*. It highlights situations where naive policy gradient updates fail, and presents provably efficient variants such as the *Conservative Greedy Algorithm* and *Trust Region Policy Optimization (TRPO)*.

1. When Policy Gradients Fail

Example 1 – Exponential Trajectories. In a chain MDP where each state $s \in \{n, \ldots, 1\}$ has two forward actions and one backward action, the uniform policy's probability of reaching the rewarding state 1 decays exponentially with n. Hence, the expected gradient magnitude

$$\|\nabla_\theta \rho(\pi)\| \leq \frac{n}{1-\gamma}\left(\frac{1}{2}\right)^{n/2},$$

implying exponentially many samples are needed to obtain a nonzero gradient estimate.

Example 2 – Two-State Trap. Even with two states (i, j), policy gradients may worsen the policy. Because gradient updates scale with the stationary distribution $d^\pi(s)$, the high-probability state i dominates learning. The update

$$\theta_{s,a} \leftarrow \theta_{s,a} + \alpha\, d^\pi(s)\pi_\theta(s,a)\big(Q^\pi(s,a) - V^\pi(s)\big)$$

can increase the probability of looping on i, worsening performance before eventual recovery in exponential time.

2. Performance Difference Lemma

A key result that underlies all policy improvement analysis:

$$\rho(\tilde{\pi}) - \rho(\pi) = \frac{1}{1-\gamma}\,\mathbb{E}_{s\sim(1-\gamma)d^{\tilde{\pi}},\,a\sim\tilde{\pi}}[A^{\pi}(s,a)], \quad A^{\pi}(s,a) = Q^{\pi}(s,a) - V^{\pi}(s).$$

This identity expresses the performance gap between any two policies in terms of the *advantage* of π under the new policy's state distribution.

3. Conservative Greedy Policy Improvement

To ensure monotonic improvement, *Kakade & Langford (2002)* propose the update:

$$\pi_{\text{new}}(s,a) = (1-\alpha)\pi(s,a) + \alpha\pi'(s,a),$$

where π' is a candidate (greedy) policy and α controls conservativeness.

Guaranteed Improvement. For any π',

$$\rho(\pi_{\text{new}}) - \rho(\pi) \geq \alpha\, A_{\pi}(\pi') - \frac{2\alpha^2\gamma\epsilon}{1-\gamma(1-\alpha)},$$

where

$$A_{\pi}(\pi') = \tfrac{1}{1-\gamma}\mathbb{E}_{s,a\sim d^{\pi},\pi'}[A_{\pi}(s,a)], \quad \epsilon = \tfrac{1}{1-\gamma}\max_s \Big|\sum_a \pi'(s,a)A_{\pi}(s,a)\Big|.$$

Choosing Parameters. If rewards are bounded by R, selecting

$$\alpha = \frac{A_{\pi}(\pi')(1-\gamma)^3}{4R}$$

yields the provable bound

$$\rho(\pi_{\text{new}}) - \rho(\pi) \geq \frac{(1-\gamma)^3 A_{\pi}(\pi')^2}{8R}.$$

Thus each update is guaranteed non-negative improvement.

Termination Guarantee. The conservative greedy algorithm stops after at most

$$O\left(\frac{R^2}{\delta^2}\right)$$

iterations to find a π satisfying

$$(1-\gamma)\max_{\pi'} A_\pi(\pi') \leq \delta.$$

4. Local vs. Global Optimality

Even a locally optimal policy can differ significantly from the global optimum if state distributions differ.

$$\rho(\pi^*;\mu^*) - \rho(\pi;\mu^*) \leq \frac{\delta}{(1-\gamma)^2}\left\|\frac{d_{\pi^*,\mu^*}}{d_{\pi,\mu}}\right\|_\infty.$$

Choosing a more uniform initial state distribution mitigates this gap.

5. Trust Region Policy Optimization (TRPO)

Schulman et al. (2015) generalized the conservative update by replacing the mixture constraint with a *trust region* constraint based on total-variation or KL divergence:

$$\rho(\tilde{\pi}) - \rho(\pi) \geq A_\pi(\tilde{\pi}) - \frac{2\gamma\epsilon}{1-\gamma}\,D_{TV}^{\max}(\pi,\tilde{\pi})^2 \geq A_\pi(\tilde{\pi}) - \frac{4\gamma\epsilon}{1-\gamma}\,D_{KL}^{\max}(\pi,\tilde{\pi}).$$

This ensures safe updates without forming explicit policy mixtures.

TRPO Objective. Find the next policy parameter $\tilde{\theta}$ by solving

$$\max_{\tilde{\theta}}\ A_\theta(\tilde{\theta}) \quad \text{s.t.} \quad D_{KL}^{\max}\big(\pi_\theta \,\|\, \pi_{\tilde{\theta}}\big) \leq \eta,$$

where η controls the trust-region size.

6. Key Takeaways

- Naive policy gradients can converge exponentially slowly or even worsen performance.

- The **Performance Difference Lemma** formalizes exact performance change between policies.
- **Conservative Greedy Algorithm** ensures monotonic improvement via step-size control.
- **TRPO** removes mixtures by constraining updates through KL or TV divergence.
- Both frameworks form the theoretical foundation for modern stable policy-gradient methods (e.g., PPO).